Sandbox Mini-App Approach

CodeFixPanda_Kotlin · senior Android interview preparation · 2026-11-11 · build 64.3

Purpose

The real app is large. To absorb it, each concept from the code walkthrough gets a tiny, runnable example in a separate sandbox Activity, `MainActivity2`. The real code stays untouched, and the sandbox is never merged into `develop`.

Branches

These follow the branch conventions document. Both branches were cut from `develop` at 95ed21d, build 63.9, last merged PR #61.

Branch	Role	Merge?
`feature-set/codeWalkthrough`	Staging. Stays at the `develop` tip. It moves up to the step's head only when Gene says so.	No PR is opened
`sandbox/miniAppDemos`	Step branch. Every commit is made here, and we stay on it until Gene says otherwise. It is named `sandbox/` instead of `refinement/` because it is not a refinement.	Never merged

The earlier branches `sandbox/walkthrough-mini-app`, `sandbox/walkthrough-step-02-koin-lazy-single` and the session branch `claude/festive-mayer-purx3h` and the first step branch `refinement/codeWalkthrough-koin` are abandoned. Gene deletes them.

What the step branch changes

- **New file** `app/src/main/java/io/github/genesjc/codefixpanda/MainActivity2.kt`, the mini app.
- **Edited** `app/src/main/AndroidManifest.xml`. The launcher `<intent-filter>` moved from `MainActivity` to `MainActivity2`. `MainActivity` is otherwise unchanged.
- **Edited** `app/build.gradle.kts`. The build number is 64.n, one step per commit.
- **Docs** in `_meta/docs`: the conventions PDF and this report.
- `CodeFixPandaApplication` still runs for both Activities, so Koin is started and the sandbox can use the real dependency injection setup.

Switching between the sandbox and the real app

Move this block between the two `<activity>` elements in the manifest, then rebuild:

```
<intent-filter>
    <action android:name="android.intent.action.MAIN" />
    <category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
```

Under `MainActivity` the real app runs. Under `MainActivity2` the sandbox runs.

Working method for each chunk

1. Walk one chunk of real code, following the conventions PDF.
2. Where it helps, add a minimal example of the same concept to `MainActivity2.kt` as one commit on the step branch, with the build number raised and shown at the end of the commit subject.
3. Run it and compare what happens with what the real code does.

Progress

Chunk	Real code	Concept	Sandbox demo
1	`MainActivity.onCreate`	Activity, `enableEdgeToEdge`, `setContent`	Not yet
2	`CodeFixPandaApplication.onCreate`	Koin starts once per process, and a `single` is built lazily	Committed in 64.1, not yet run

Next: Chunk 3, `CodeFixPandaApp()` in `ui/CodeFixPandaApp.kt`, the root composable that `setContent` calls.

The Chunk 2 demo

It registers a tiny Koin module and logs under the tag `Mini`. The expected Logcat order is `module registered, Greeter not built yet`, then `Greeter constructed`. The second line appears at first composition, after `onCreate` returns, when the `setContent` lambda first reads the injected value. That shows `modules(appModule)` registers definitions without building them.

Known limits

- **Not compiled yet.** The session container has no Android SDK, so the demo has not been built or run. The manifest is valid XML. The expected Logcat order is what the code should produce, not an observed result.
- CI runs only for pushes and PRs on `develop` and `main`, so these branches get no CI run. The blank-line guard rules were applied by hand.
- Because the manifest is edited to switch Activities, move the launcher filter back before running the real app.

For another conversation joining this work

1. Read `_meta/docs/code-walkthrough-conventions.pdf` and the branch conventions document.
2. Check out `sandbox/miniAppDemos`.
3. Follow the conventions. Do not merge the step branch or open a PR from the staging branch.